

US100 MODELLING STUDY

Subject: Free-float index construction, passive ownership and rebalance-flow analysis

Executive Summary

Question	Result
1. Tracking AUM	\$940.19bn
2. RTX weight	0.6276%
3. GOOG passive holding	\$26.80bn; 11,889,500 shares
4. ORCL weight at 80.3% float	0.6621%
5. Largest AAPL deletion flow	-\$91.69bn; -663,130,520 shares
6a. Largest flow pressure	AAPL: 13.28x 30-day ADV
6b. AMZN 20-for-1 split	Shares x20; price /20; value and weight unchanged

Core Method

- Float shares = shares outstanding x free float
- Float market cap = price x float shares
- Index weight = security float market cap / total index float market cap
- Tracking AUM = tracking factor x total index float market cap
- Passive shares held = tracking factor x float shares

Important correction to the brief: index weight cannot equal float market cap divided by tracking AUM, because that would make weights sum to 1 / 4.3% rather than 100%. The model uses the standard index-weight definition and uses tracking AUM only to convert weight changes into dollar flows.

Assumptions

- Tracking factor is fixed at 4.3%.
- For the AAPL deletion, total tracking AUM is held constant at the event instant.
- Flow sign is new holding minus old holding; negative values are sells.
- Prices, shares and floats are taken directly from the supplied hypothetical constituent file.

Source: Intropic Tracking Guide; supplied US100 constituent data.

1. We estimate the tracking of the US100 index to be 4.3%. What is the total amount of money in terms of dollars (AUM) that is tracking the index?

Answer: approximately \$940.19bn.

- Total float market cap = \$21,864,916,552,920.
- Tracking AUM = \$21,864,916,552,920 x 4.3% = \$940,191,411,775.56 (approximately \$940.19bn).

2. What is the current weight of RTX in the index?

Answer: 0.6276%.

- RTX float market cap = \$137,227,355,280.00.
- RTX weight = \$137,227,355,280.00 / \$21,864,916,552,920 = 0.6276%.

3. How much of GOOG stock is currently held by passive funds tracking the index?

- In dollar value?
- In shares amount?

Answer: approximately \$26.80bn, equal to 11,889,500 shares.

- GOOG float shares = 316,000,000 x 87.5% = 276,500,000.
- Passive shares = 276,500,000 x 4.3% = 11,889,500.
- Passive value = 11,889,500 x \$2,253.69 = \$26,795,247,255.00.

4. What would the new weight of ORCL be in the index if their float was increased by 20% to 80.3%?

Answer: 0.6621%, up from 0.4980%.

- New ORCL float market cap = \$67.63 x 2,670,000,000 x 80.3% = \$144,999,396,300.00.
- The index denominator must also be updated: old total - old ORCL float cap + new ORCL float cap.
- New ORCL weight = \$144,999,396,300.00 / \$21,901,030,972,920.00 = 0.6621%.

Key Python

```
df["float_shares"] = df.shares_outstanding * df.free_float
df["float_market_cap"] = df.price_usd * df.float_shares
total_float_cap = df.float_market_cap.sum()
df["weight"] = df.float_market_cap / total_float_cap
tracking_aum = total_float_cap * 0.043
```

Q5. In a shock move AAPL are to be deleted from the index. What would be the largest flows resulting from this deletion (there is no replacement company)? Here: $\text{flow} = \text{current_weight_in_index} - \text{previous_weight_in_index}$

a) Top 10 flows based on USD in absolute terms

In a shock move, AAPL is deleted with no replacement. What are the top 10 absolute flows?

AAPL is the largest flow: approximately -\$91.69bn (-663,130,520 shares). The sale proceeds are redistributed pro rata across the remaining 99 constituents.

Ticker	Old wt.	New wt.	Flow USD	Flow shares
AAPL	9.7524%	0.0000%	\$-91,691,057,000	-663,130,520
MSFT	8.9172%	9.8808%	\$9,059,801,132	34,998,845
AMZN	4.4606%	4.9426%	\$4,531,913,987	40,305,176
GOOGL	3.0820%	3.4150%	\$3,131,281,602	1,394,880
GOOG	2.8500%	3.1580%	\$2,895,561,009	1,284,809
TSLA	2.6813%	2.9711%	\$2,724,196,660	3,862,958
BRK/B	2.2403%	2.4824%	\$2,276,157,121	8,508,362
JNJ	2.1613%	2.3949%	\$2,195,892,873	12,222,492
UNH	2.1493%	2.3815%	\$2,183,633,158	4,368,927
NVDA	1.7738%	1.9655%	\$1,802,182,185	11,107,440

Top 10 absolute flows after AAPL deletion



Cash conservation: purchases and AAPL sale proceeds both round to \$91.69bn. Net flow rounds to zero; the unrounded residual is below one cent and arises from floating-point arithmetic.

Interpretation: AAPL dominates the notional flow. Among the additions, MSFT and AMZN receive the largest dollar purchases because the remaining index is reweighted proportionally to existing float capitalisation.

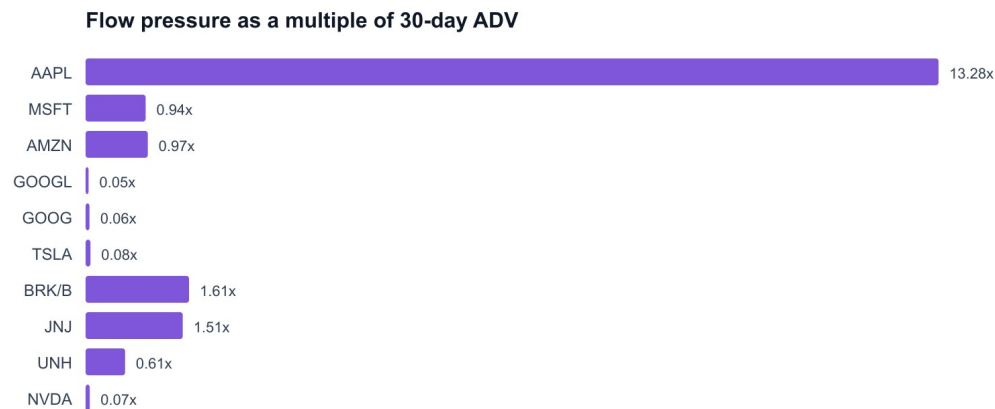
Q6. Trading by a passive fund can impact a stock's price by exerting strong buying/selling pressure at the close of business on trade date. To proxy this price impact we look at the flows in terms of the average daily volume (ADV) of a stock. E.g. if we expect 10m shares of company B to be sold by passive funds and that company's ADV is 2m shares, then we would say that the trade was equivalent to 5 ADV (10/2)

a) Find a data source for average daily volume and use it to calculate the ADV of the above 10 flows.

ADV source: Yahoo Finance daily share volume. ADV is the arithmetic mean of 30 trading sessions from 7 May 2026 to 18 June 2026, retrieved 21 June 2026.

The case does not specify an event date. Therefore, this is a clearly timestamped current liquidity proxy. For a live forecast, the 30-session window should end immediately before the rebalance trade date.

Ticker	Abs. flow shares	30-day ADV	Flow / ADV
AAPL	663,130,520	49,931,193	13.28x
MSFT	34,998,845	37,415,287	0.94x
AMZN	40,305,176	41,622,950	0.97x
GOOGL	1,394,880	30,767,840	0.05x
GOOG	1,284,809	21,491,810	0.06x
TSLA	3,862,958	50,621,557	0.08x
BRK/B	8,508,362	5,283,960	1.61x
JNJ	12,222,492	8,078,033	1.51x
UNH	4,368,927	7,104,010	0.61x
NVDA	11,107,440	170,728,257	0.07x



Answer: AAPL is the largest liquidity event at 13.28x ADV. BRK/B (1.61x) and JNJ (1.51x) also exceed one day of normal volume. They are the strongest candidates for closing-auction price pressure.

Caveat: ADV is a liquidity proxy, not a price-impact model. Auction depth, event anticipation, active investors, market volatility and dealer participation can materially change realised impact.

b) What would be the effect of a 20:1 stock split in AMZN stock on this passive fund's holdings?

Before: 372,980,280 shares at \$112.44; value approximately \$41.94bn.

After: 7,459,605,600 shares at \$5.6220; value remains \$41.94bn.

The split multiplies shares by 20 and divides price by 20. Float market cap, index weight and passive dollar exposure are unchanged. No rebalance trade is required solely because of the split.

Tracking-Factor Sensitivity

Weights do not depend on the tracking assumption. Tracking AUM and dollar flows scale linearly with it, so the 4.3% estimate is the largest commercial uncertainty in the flow forecast.

Tracking factor	Tracking AUM	AAPL deletion sell
3.0%	\$655.95bn	\$-63.97bn
4.3%	\$940.19bn	\$-91.69bn
5.0%	\$1,093.25bn	\$-106.62bn
7.0%	\$1,530.54bn	\$-149.26bn

Validation Controls

Control	Result
100 constituents loaded	PASS
Pre-event weights sum to 100%	PASS (100.000000%)
Post-deletion weights sum to 100%	PASS (100.000000%)
Purchases equal AAPL sale proceeds	PASS
Total net flow equals zero	PASS
AMZN pre/post-split value unchanged	PASS

These controls are designed to fail loudly if a formula, denominator or sign convention is changed incorrectly.

Technical Appendix

Core Rebalance Logic

```
remaining = df[df["ticker"] != "AAPL"].copy()
remaining["new_weight"] = (
    remaining["float_market_cap"]
    / remaining["float_market_cap"].sum()
)
remaining["weight_change"] = (
    remaining["new_weight"] - remaining["weight"]
)
remaining["flow_usd"] = remaining["weight_change"] * tracking_aum
remaining["flow_shares"] = remaining["flow_usd"] / remaining["price_usd"]

# Include the deleted security in the absolute-flow ranking.
aapl_flow_usd = -aapl["weight"] * tracking_aum

assert abs(all_flows["flow_usd"].sum()) < 0.01
assert abs(remaining["new_weight"].sum() - 1) < 1e-12
```

ADV Calculation

```
# 30 most recent valid trading sessions in the chosen window
adv_30d = sum(volumes[-30:]) / 30
flow_adv = abs(flow_shares) / adv_30d
```

Reproducibility

The submitted notebook contains all calculations, visible outputs, cached ADV observations, a parameterised Yahoo ADV refresh function, charts, sensitivity analysis and validation assertions. It runs offline with the supplied CSV and the pinned pandas dependency; internet access is required only to refresh ADV.

Sources

1. Intropic, Tracking Guide, supplied PDF and Notion reference.
2. FTSE Russell, Russell US Equity Indexes Construction and Methodology, supplied PDF.
3. FTSE Russell, Corporate Actions and Events Guide for Market Capitalisation Weighted Indices, supplied PDF.
4. Yahoo Finance chart endpoint, daily share volume for AAPL, MSFT, AMZN, GOOGL, GOOG, TSLA, BRK-B, JNJ, UNH and NVDA; 30 sessions from 7 May 2026 to 18 June 2026; retrieved 21 June 2026.
5. Supplied hypothetical constituent file, US100_constituents (1) (1).csv.

Yahoo Finance is used because the case does not provide institutional market data access. In production, the same calculation should be refreshed from Bloomberg, LSEG or another licensed source immediately before the event.